OTH REGENSBURG

Praktikum Regelungstechnik (RTP)

Versuch T4:
RetBot system identification, part II

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Gruppe 38
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1 Preparations

1.1. (aV) Steps to solve the control task

In order to solve a control problem, a mathematical model of the controlled system is first developed and the performance requirements of the control system are defined. The controller is then designed based on this, during which the structure and parameters of the control law are specified. Finally, the controller is implemented in hardware or software and validated on the actual system. This process is illustrated in Figure 1.1 according to Lunze [1].

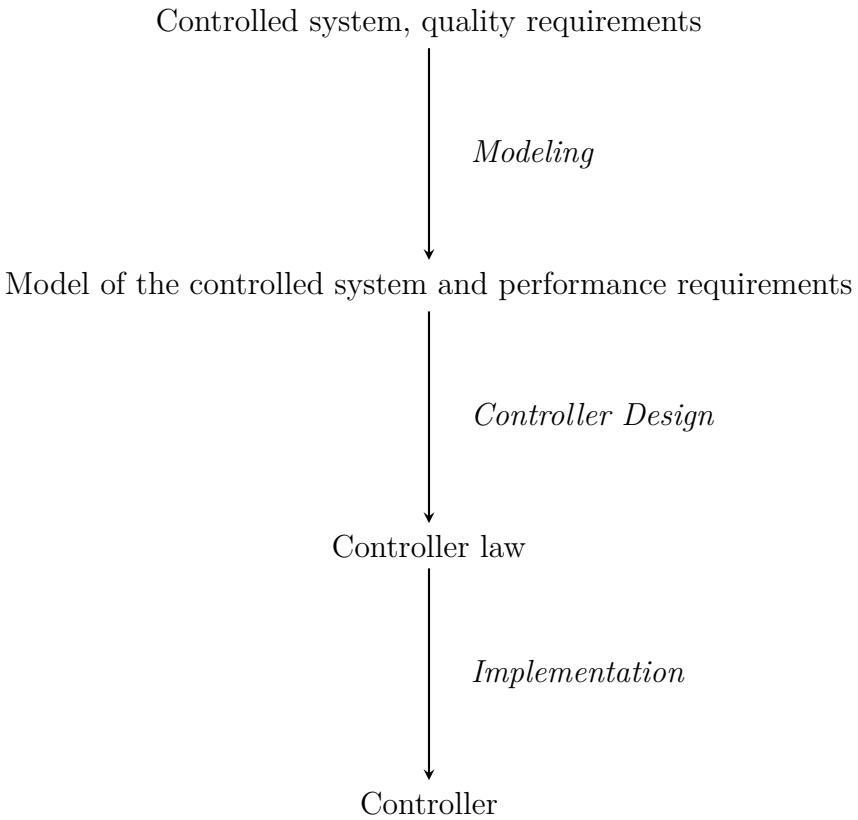


Abbildung 1.1.: Approach to control engineering problems according to Lunze [1]

1.2. (bV) Block diagram of the control loop

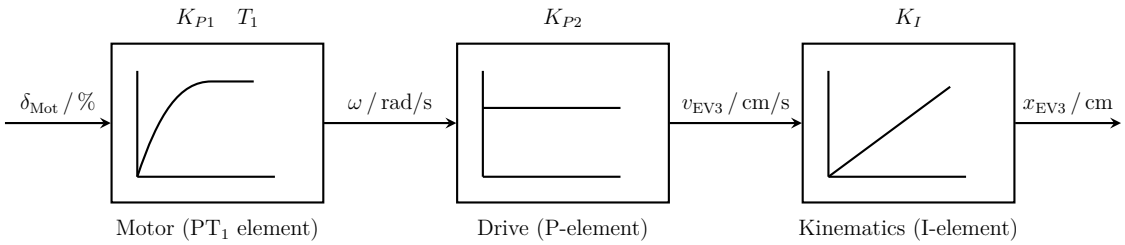
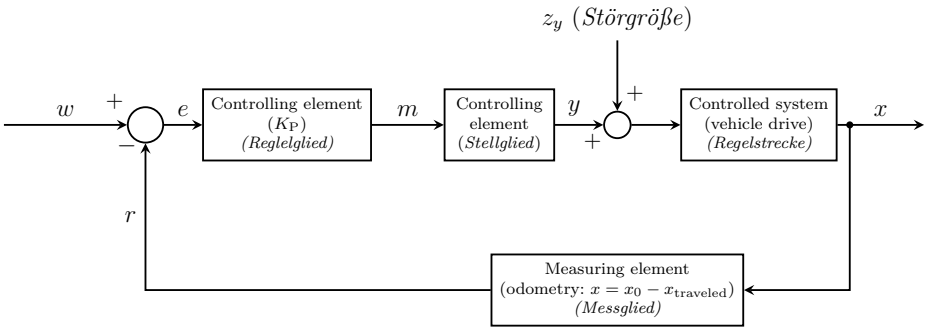


Abbildung 1.2.: The block diagram illustrates the configuration of the EV3 robot.

The control process of the EV3 robot (Figure 1.2) begins with the duty cycle δ, which is provided as an input signal in the form of pulse width modulation (PWM). This signal determines the average voltage of the DC motor and, consequently, the torque and rotational speed. The resulting rotational speed is transmitted to the wheels via the gearbox and converted into the robot's linear translational speed. Finally, the robot's kinematics describe the relationship between the rotational movements of the individual wheels and the resulting overall movement of the robot.

1.3. (cV) Block diagram of a distance control system (according to DIN IEC 60050-351).

Diagram 1.3 shows a block diagram of a distance control system using a P controller according to DIN IEC 60050-351. The system consists of a reference variable w (target distance), a control difference e, a controller output m, a manipulated variable y, a disturbance variable z_y, a controlled variable x (actual distance), and a feedback variable r. The controlling element (K_{P}) processes the control difference to generate the controller output, which the final controlling element converts into the manipulated variable acting on the controlled system. The disturbance z_y acts additively on the manipulated variable, and the measuring element derives the actual distance from the distance travelled, $x = x_0 - x_{\mathrm{traveled}}$, providing the feedback variable r.



w: reference variable (*Führungsgröße*, target distance)
e: control difference (*Regeldifferenz*), $e = w - r$
m: controller output (*Reglerausgangsgröße*)
y: manipulated variable (*Stellgröße*)
z_y: disturbance variable (*Störgröße*)
x: controlled variable (*Regelgröße*, actual distance)
r: feedback variable (*Rückführgröße*)

Abbildung 1.3.: Block diagram of a distance control system using a P controller according to DIN IEC 60050-351

2 Experiment T4: RetBot system identification, part 2

2.1. (d) System description

2.2. (e)

2.3. (f)

A Appendix Frontmatter

A.1. Declarations

1. Wir erkläre hiermit, dass wir dieses Dokument selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate als solche gekennzeichnet haben.

Regensburg, den 2026-05-01,

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Ort, Datum und Unterschrift

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B Appendix Code and Resources

B.1. Code used for the analysis of the Preperations

Editorial note — Source code omitted

The full source code for the analysis (a Jupyter notebook of this time approximately 1,500 lines) is *not* included in this appendix. Including it as a verbatim listing caused major build instability again and was therefore deliberately omitted.
The complete source can be obtained by either of the following means:

- By email request to `janis.graf@st.oth-regensburg.de`
- Via requesting access to the GitHub repository: `https://github.com/janisgra/RTP-T3-SoSe2026`

Literatur

[1] Jan Lunze. *Regelungstechnik 1: Systemtheoretische Grundlagen, Analyse und Entwurf linearer Regelungssysteme*. 9. Aufl. Berlin, Heidelberg: Springer Vieweg, 2016.